

# Monopile Installation Record HB-A10

## Harrow Bay Offshore Wind · Foundations

Elena Márquez · 03 July 2026 · HB-CON-IR-A10

| Position | Date        | Vessel       | Contract               |
|----------|-------------|--------------|------------------------|
| HB-A10   | 24 Jun 2026 | MV Sea Atlas | WE-CON-2025-040 / -027 |

### 1. Position and pile data

| Item                   | Design                | As-installed           |
|------------------------|-----------------------|------------------------|
| Coordinates (WGS84)    | 53.93090 N, 1.18220 E | within 19 cm of target |
| Water depth (m LAT)    | 33.4                  | 33.4                   |
| Pile length (m)        | 76.9                  | 76.9                   |
| Pile mass (t)          | 1,357                 | –                      |
| Target penetration (m) | 34.7                  | 35.0                   |
| Verticality            | ≤ 0.25°               | 0.18°                  |

### 2. Driving record

| Item                    | Value      |
|-------------------------|------------|
| Start of soft start     | 05:11      |
| Driving duration        | 4 h 15 min |
| Total blow count        | 5,287      |
| Maximum hammer energy   | 2,517 kJ   |
| Self-weight penetration | 5.0 m      |
| Interruptions           | None       |

### 3. Noise and marine mammal mitigation

| Item                          | Value                                                 |
|-------------------------------|-------------------------------------------------------|
| Bubble curtain                | Double big bubble curtain, both rings operational     |
| Peak SEL at 750 m             | 158.2 dB (limit 160 dB)                               |
| Pre-piling search (MMO + PAM) | 30 min, marine mammal detected – start delayed 15 min |
| Acoustic deterrent device     | Activated 30 min before soft start                    |

### 4. Transition piece

| Item | Value |
|------|-------|
|------|-------|

|                                    |                       |
|------------------------------------|-----------------------|
|                                    |                       |
| TP installed                       | 02 Jul 2026           |
| Flange flatness after installation | 0.5 mm (limit 2.0 mm) |
| Bolt tensioning                    | 148 × M72 – completed |
| TP level                           | +0.03° (limit 0.25°)  |

5. Non-conformances

NCR-HB-044: minor coating damage on pile top during upending, repaired offshore.

6. Sign-off

| Role                             | Name                     | Date        |
|----------------------------------|--------------------------|-------------|
| Offshore Installation Supervisor | Atlas Marine Contractors | 25 Jun 2026 |
| Client Representative            | Felix Brandt             | 25 Jun 2026 |
| Reviewed                         | Elena Márquez            | 26 Jun 2026 |